Quasi-Poisson models

Last time: quasi-Poisson models

Poisson:

- Mean = λ_i
- Variance = λ_i

quasi-Poisson:

- Mean = λ_i
- Variance = $\phi\lambda_i$
- Variance is a linear function of the mean

ϕ: dispersion parameter

In both cases: same systematic component

$$\log(\lambda_i) = \beta_0 + \beta_1 x_{i1} + \cdots$$

$$SE(\hat{\beta})_{QP} = \sqrt{\hat{\phi}}\, SE(\hat{\beta})_{Poisson}$$

Class activity

Work on the class activity (handout).

Class activity

pchisq(11540, df = 2004, lower.tail=F) ≈ 0



> Which model do you think is more appropriate for the data?

Model 2 (Quasi-Poisson)

- $\hat{\phi} = 5.52$ (variance > 5 × mean)
- GoF test

H_0: Model 1 is a good fit

H_A: Model 1 is not a good fit

Test statistic: residual dev = 11540

df = 2004

If H_0 is true, then

residual dev $\approx \chi^2_{2004}$

11540 >> 2004

$\Rightarrow$ p-value ≈ 0 (reject H_0)

$H_0: \beta_{education2} = 0$

$H_A: \beta_{education2} \neq 0$

Class activity

Is there a difference in the average number of cigarettes smoked between smokers with some college education and smokers with no college education, after accounting for sex, age, diabetes, and BMI?

$$t = \frac{0.01646 - 0}{(\sqrt{5.52})(0.01263)} = 0.557$$

Model 1:

```
                Estimate Std. Error z value Pr(>|z|)
...
education2    0.0164626  0.0126312   1.303 0.192462
...
```

$0.01646 \pm t^* (\sqrt{5.52})(0.01263)$

Model 2: check whether CI contains 0 (or exponentiate & check whether contains 1)

```
(Dispersion parameter for quasipoisson family taken
    to be 5.519388)
```

Class activity

Work on the class activity (course website).

Class activity

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.8371917  0.0489922  57.911  < 2e-16 ***
male         0.4561700  0.0111074  41.069  < 2e-16 ***
age         -0.0068063  0.0006789 -10.026  < 2e-16 ***
education2   0.0164626  0.0126312   1.303 0.192462
education3   0.0164274  0.0160498   1.024 0.306057
education4  -0.0150328  0.0172291  -0.873 0.382921
diabetes    -0.0253978  0.0394181  -0.644 0.519368
BMI          0.0050011  0.0014065   3.556 0.000377 ***
```

Research question: Is there a difference in the average number of cigarettes smoked between smokers with some college education and smokers with a college degree, after accounting for sex, age, diabetes, and BMI?

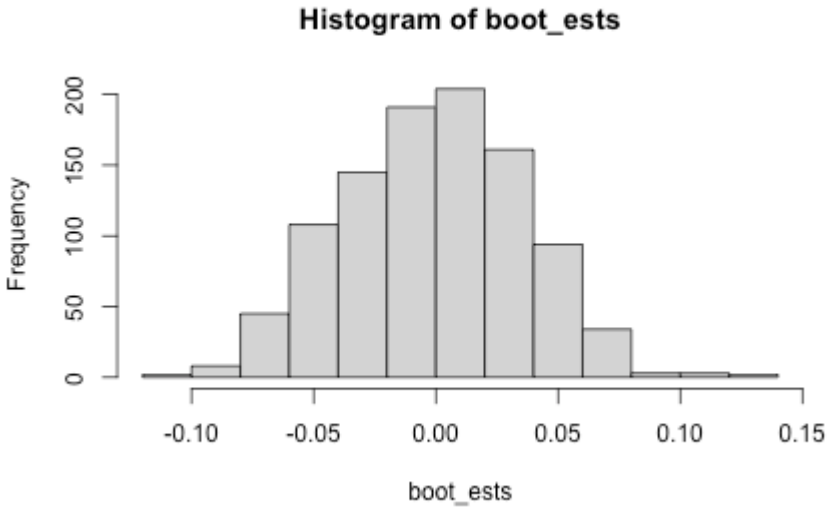
> Can I calculate a confidence interval to answer this question, using just the R output shown here?

Class activity

Bootstrap resampling to compute the confidence interval:

Class activity

```
hist(boot_ests)
```




```
sd(boot_ests)
```

```
## [1] 0.03682651
```

Another question

```
m2 <- glm(cigsPerDay ~ male + age + education + diabetes + BMI,
          data = smokers, family = quasipoisson)
summary(m2)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.837192   0.115099  24.650  < 2e-16 ***
male         0.456170   0.026095  17.481  < 2e-16 ***
age         -0.006806   0.001595  -4.267 2.07e-05 ***
education2   0.016463   0.029675   0.555    0.579
education3   0.016427   0.037706   0.436    0.663
education4  -0.015033   0.040477  -0.371    0.710
diabetes    -0.025398   0.092606  -0.274    0.784
BMI          0.005001   0.003304   1.513    0.130
```

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

Nested tests with quasi-Poisson models

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

- In Poisson regression, we would use a likelihood ratio test
- However, the quasi-Poisson model includes the estimated dispersion, $\widehat{\phi}$. We need to use an **F-test** instead

Nested tests with quasi-Poisson models

```
m1 <- glm(cigsPerDay ~ male + age + education +
            diabetes + BMI,
          data = smokers, family = quasipoisson)
m2 <- glm(cigsPerDay ~ male + age + diabetes + BMI,
          data = smokers, family = quasipoisson)

m1$deviance
```

```
## [1] 11539.56
```

```
m2$deviance
```

```
## [1] 11543.84
```

```
m1$df.residual
```

```
## [1] 2004
```

```
m2$df.residual
```

```
## [1] 2007
```

Nested tests with quasi-Poisson models

```
m1$deviance
```

```
## [1] 11539.56
```

```
m2$deviance
```

```
## [1] 11543.84
```

```
pf(0.258, 3, 2004, lower.tail=F)
```

```
## [1] 0.8556648
```

Limitations of quasi-Poisson models

Important: quasi-Poisson models do *not* describe a full distribution for the response (there is no "quasi-Poisson" distribution). They describe *only* the mean and variance of the response.

As a result, one cannot calculate probabilities for a quasi-Poisson:

```
m2 <- glm(cigsPerDay ~ male + age + diabetes + BMI,
          data = smokers, family = quasipoisson)
rootogram(m2)
```

```
Error in prodist.glm(object) :
family is not associated with a full probability distributi
```